

Renal Transplant

ORG: S-1015 (ISC)

[Link to Codes](#)**MCG Health**Inpatient & Surgical
Care
28th Edition

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4)(5):[\[A\]](#)
 - ☐ End-stage renal disease, as indicated by **1 or more** of the following(1):
 - Need for chronic renal replacement therapy due to irreversible renal disease
 - Glomerular filtration rate 30 mL/min/1.73m² (0.50 mL/sec/1.73m²) or less  eGFR - Adult Calculator and irreversible renal disease
 - ☐ No contraindications present, as indicated by **ALL** of the following(1):
 - No multiple myeloma or monoclonal immunoglobulin deposition disease (light or heavy chain deposition) unless in stable remission
 - No amyloidosis with extensive extrarenal involvement (eg, cardiac amyloidosis)
 - No decompensated cirrhosis unless candidate for combined liver-kidney transplant
 - No severe irreversible obstructive or restrictive lung disease (eg, oxygen-dependent irreversible disease)
 - No severe uncorrectable symptomatic heart disease (eg, New York Heart Association functional class IV heart failure, coronary artery disease, valvular disease) unless candidate for combined heart-kidney transplant or evaluated by a transplant cardiologist
 - No progressive central neurodegenerative disease (eg, Alzheimer, Parkinson, Huntington diseases)
 - No unstable psychiatric disorder that affects decision making
 - No ongoing substance use disorder that affects decision making
 - No inability to comply with ongoing medical care
 - No active bacterial, viral, parasitic, or fungal infection other than hepatitis C
 - No stroke within the last 6 months or transient ischemic attack within the last 3 months
 - No active gastrointestinal disease including symptomatic peptic ulcers, diverticulitis, inflammatory bowel disease, or gallbladder disease
 - No acute pancreatitis within the last 3 months
 - No acute hepatitis (eg, elevated INR or transaminitis)
 - No active malignancies[A]

- No surgical contraindications (eg, urologic or vascular problems)

Alternatives to Procedure

- Alternatives include(6)(7):
 - Hemodialysis
 - Peritoneal dialysis

Operative Status Criteria

Goal Length of Stay: 4 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(1)(2)(3)(4)(6)(8)(9):
 - Preoperative evaluation, including:
 - Multidisciplinary renal transplant team evaluation for:
 - Etiology of end-stage renal disease
 - Alternatives to renal transplant
 - Contraindications to renal transplant
 - Risk of graft failure
 - Psychosocial evaluation(10)
 - Assessment of medical risk[B](11)(12)
 - Protocol as specified by transplant center
 - Education of patient and dialysis unit staff
 - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool [SR](#).
 - Diagnostic test scheduling, including(1):
 - Laboratory testing
 - Glucose tolerance test for patient not known to be diabetic
 - Infectious disease screening, including latent TB, hepatitis B and C, cytomegalovirus, varicella, MMR, HIV with viral load and CD4 count if HIV positive, and antibody titers
 - Immunology screening, including blood and tissue typing and HLA antibodies
 - Nutrition evaluation(13)
 - Dental evaluation
 - Chest x-ray
 - Renal imaging (renal ultrasound, CT scan, or MRI)
 - Echocardiogram(11)
 - Cancer screening as appropriate for age, gender, and individual risk factors
 - Urodynamic evaluation (eg, voiding dysfunction, neurogenic bladder, suspected noncompliant bladder)(14)
 - Preoperative treatment, procedures, and stabilization, including:
 - Immunization vaccinations as appropriate (eg, hepatitis, varicella-zoster, pneumococcus, influenza, COVID-19)(15)
 - Desensitization for live-donor recipients with high HLA sensitivity, HLA mismatch, or ABO mismatch(16)(17)(18)(19)
 - Bladder augmentation or urinary diversion procedures(14)
 - Stress echocardiogram (eg, for echocardiogram or ECG findings suggestive of ischemic heart disease, or history of known coronary heart disease without recent imaging)(11)
 - Coronary artery imaging (eg, CT angiogram or invasive coronary angiography)(11)
 - Smoking cessation counseling(1)
 - Alcohol cessation counseling
 - Weight loss counseling(1)
 - Physical rehabilitation for frail patients(1)
 - Pretransplant cystoscopy if high risk for bladder cancer (eg, heavy smoking, cyclophosphamide therapy)
 - Pretransplant native kidney nephrectomy[C](1)
 - Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
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1	• OR to ICU • Social Determinants of Health Assessment • Readmission Risk Assessment • Discharge planning	• Clinical Indications met[D]	• Bed rest or up at bedside	• IV fluids and medications • Oral fluids	• Dialysis • Central lines • Urethral catheter • CBC • Electrolytes and renal function tests	• Immunosuppressive medication and corticosteroids • Multimodal analgesia, possible PCA • Perioperative antibiotics
2	• Floor • Social Determinants of Health Assessment • Readmission Risk Assessment	• Hemodynamic stability	• Out of bed • Possible early ambulation	• IV fluids and medications • Advance diet as tolerated • Oral medications	• Central lines absent • CBC • Electrolytes and renal function tests	• Immunosuppressive medication and corticosteroids • Multimodal analgesia • Antibiotics absent
3	• Floor • Social Determinants of Health Assessment • Readmission Risk Assessment	• Renal function adequate • No evidence of ileus or bowel obstruction	• Ambulatory	• Oral fluids tolerated • IV or oral medications	• Urethral catheter absent • CBC • Electrolytes and renal function tests • Immunosuppressive medication monitoring	• Immunosuppressive medication and corticosteroids • Multimodal analgesia
4	• Floor • Social Determinants of Health Assessment • Readmission Risk Assessment	• Pain absent or managed • No evidence of postoperative or surgical site infection	• Ambulatory	• Oral hydration • Oral medications	• Renal scan • Immunosuppressive medication monitoring • CBC • Electrolytes and renal function tests	• PCA absent[E] • Immunosuppressive medication and corticosteroids • Begin oral corticosteroid taper
5	• Social Determinants of Health Assessment • Readmission Risk Assessment • Floor to discharge[F] • Complete discharge planning	• No evidence of graft rejection • No evidence of postoperative or surgical site infection • Renal function adequate • Diet tolerated • Pain absent or managed • Discharge plans and education understood	• Ambulatory or acceptable for next level of care	• Oral hydration[G] • Oral medications or regimen acceptable for next level of care • Oral diet or acceptable for next level of care	• Immunosuppressive medication monitoring • CBC • Electrolytes and renal function tests	• Oral immunosuppressive medication and corticosteroids

(3)(6)(9)(20)(21)(22)(23) Recovery Milestones are indicated in **bold**.**Goal Length of Stay: 4 days postoperative**

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(3)(6)(9)(23):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Complications of procedure(4)(8)
 - Delayed graft function may require adjustment of immunosuppression, hemodialysis, Doppler ultrasound, renogram, or biopsy.(24)(25)
 - Surgical complications may require renal imaging procedures, interventional radiology procedures, cystoscopic procedure, or reoperation; examples include(26)(27)(28)(29):
 - Urine leaks and urinomas
 - Ureteral stricture or obstruction
 - Intra-abdominal or retroperitoneal hemorrhage or hematoma
 - Lymphocele
 - Abscess
 - Renal artery stenosis
 - Renal or iliac artery thrombosis
 - Renal vein thrombosis
 - Renal arterial injury (eg, dissection, pseudoaneurysm, arteriovenous or arteriocalyceal fistula)
 - Poor graft function
 - Postoperative infection may require cultures and biomarkers, diagnostic imaging, and antibiotic therapy(30)
 - Medical complications, such as acute perioperative cardiac events, respiratory failure, or sepsis from postoperative infection, may require continued ICU care.(27)(30)(31)
 - Expect brief to moderate stay extension.
 - Acute rejection(32)(33)(34)
 - Acute rejection presents as rise in creatinine with or without reduction in urine output.
 - Anticipate serologic testing for anti-HLA antibodies, urine or serum molecular biomarker assays, or biopsy of transplanted kidney.
 - Expect treatment with immunosuppressants, immunoadsorption, antibody infusion, IV immunoglobulin, antithymocyte globulin (rATG), or plasmapheresis.
 - Expect brief stay extension.
 - Simultaneous native kidney nephrectomy^[C](9)
 - Expect brief stay extension.
 - Care for comorbidities(35)
 - Patient with complex comorbidities, such as malignant hypertension, previous stroke, poorly controlled diabetes, or active cardiac disease (eg, heart failure), may require continued acute care.
 - Patient with diabetes may require major readjustments in insulin dosing due to changing renal insulin clearance.
 - Expect brief stay extension.

See Common Complications and Conditions  ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(2)(3)(6)(9)(34):
 - Diagnostic test scheduling and completion, including:
 - Laboratory tests, including CBC, electrolytes, serum calcium, magnesium, phosphate, creatinine, and urine protein excretion
 - Renal imaging(28)(37)
 - Treatment and procedure scheduling and completion, including:
 - IV antibiotics
 - Immunosuppressive medications, antibody induction therapy, and corticosteroids(38)(39)(40)
 - Post-transplant desensitization for live-donor recipients with high HLA sensitivity, HLA mismatch, or ABO mismatch (eg, plasmapheresis, immunoadsorption, IV immunoglobulin, rituximab)(18)
 - Multimodal analgesia (20)
 - Serial postoperative CBCs and renal function tests
 - Urinary catheter care and removal(20)
 - Dialysis
 - Post-transplant oral prophylactic agents (eg, for *Pneumocystis jirovecii*, cytomegalovirus, urinary tract infection)
 - Consultation, assessment, and other services scheduling and completion, including:
 - Nephrology consultation

- Urology consultation(41)
- Nutrition consultation(42)
- Smoking cessation counseling(1)
- Identification of patient at high risk for readmission to prioritize transition and post-acute care:
 - ☐ Risk of readmission is increased by presence of **1 or more** of the following(36)(43)(44)(45)(46)(47)(48):
 - Hospitalization (nonelective) in past 6 months
 - Two or more emergency department visits in past 6 months
 - No source of outpatient care other than emergency department (eg, no primary care provider)
 - Severe care transition barriers (eg, no caregiver, homeless)
 - Delayed graft function (eg, need for dialysis within 7 days following transplant)
 - History of cerebrovascular accident (stroke)
 - Prior renal transplant (retransplant needed)
- Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [SR](#)); key items include(10):
 - Pain management, including medication by PCA. See Pain Management [SR](#) for further information.
 - Wound management, observing for healing, drainage, bleeding, and signs of infection. See Wounds Due to Surgery, Injury, or Pressure [SR](#) for further information.
 - Hemodynamic stability
 - Fluid and electrolyte balance
 - Monitoring for signs of rejection, including oliguria, anuria, fever, flank tenderness or swelling, weight gain, hypertension, decreased renal function, or malaise
 - Respiratory status
 - Neuropsychiatric problems, including delirium, delusions, emotional inappropriateness, and depression

Discharge

Discharge Planning

- Discharge planning includes[H]:
 - Assessment of needs and planning for care, including(50)(51):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(49)
 - Evaluate and address patient or caregiver preferences as indicated.
 - Identify skilled services needed at next level of care, with specific attention to:
 - Genitourinary status assessment(52)
 - Medication management, adherence instruction, and side effects assessment(30)(53)(54)
 - Nutrition and hydration management(55)
 - Ongoing education required(10)
 - Transplant graft dysfunction(56)
 - Early identification of anticipated discharge destination; options include(51)(57):
 - Home, considerations include:
 - Home safety assessment. See Home Safety Assessment [SR](#) for further information.
 - ☐ Patient safe to go home; examples include(58)(59)(60):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
 - Access to follow-up care
 - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
 - Caregiver need, ability, and availability
 - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.
 - Transitions of care plan complete, including(51)(57)(61):

- Patient and caregiver education complete. See Renal Transplant: Patient Education for Clinicians [SR](#) for further information.
 - See Teach Back Tool [SR](#) for further information.
- Medication reconciliation completion includes(53)(62):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(63)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
- Plan communicated to patient, caregiver, and all members of care team, including(64):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
- Appointments planned or scheduled, which may include:
 - Primary care provider
 - Dietitian(55)
 - Nephrologist
 - Rehabilitation therapy services
 - Specialists for management of comorbidities as needed(65)
 - Transplant center or team
 - Other
- Outpatient testing and procedure plans made, which may include:
 - Laboratory testing(30)(54)
 - Other
- Referrals made for assistance or support, which may include:
 - Community services
 - Educational program (eg, chronic condition management, self-management)(66)
 - Financial, for follow-up care, medication, and transportation
 - Self-help or support groups(67)
 - Tobacco use treatment
 - Vocational rehabilitation(68)
 - Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
 - Antiembolic or compression stockings
 - Blood pressure cuff
 - Weight scale
 - Wound care equipment and supplies(69)(70)
 - Other

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Renal Transplant guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 2 published peer reviewed articles and 3 book sections.

For end-stage renal disease, an internal medicine textbook and a specialty society guideline state that renal transplant is the definitive intervention to address end-stage renal disease.(1)(3) **(EG 2)** A specialty society guideline recommends that all patients with chronic kidney disease, defined as a glomerular filtration rate of 30 mL/min/1.73m² (0.50 mL/sec/1.73m²) or less, who are anticipated to progress to irreversible end-stage renal disease should be counseled about and considered for renal transplant.(1) **(EG 2)**

Hospitalization

Readmission risk and reduction: Analysis of 1147 renal transplants found, after multivariate adjustment, that independent risk factors for 30-day readmission included delayed graft function (need for dialysis within 7 days of transplant), repeat transplant, and history of stroke.(36) **(EG 2)**

Length of Stay

A case series including 1001 patients undergoing renal transplant found that the median length of stay was 3 days.(21) **(EG 2)** A cohort series of 1171 patients (mean age 55 years) undergoing deceased-donor kidney transplant reported a 3-day median length of stay.(22) **(EG 2)** Analysis of national hospital discharge data shows 51% of hospitalized adult patients undergoing the principal procedure of renal transplant discharged in 4 days or less.(23) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify, for a given procedure, which patient-specific factors and clinical conditions are appropriate for that procedure. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately perform a procedure, evaluating whether the potential benefits of a procedure outweigh the potential risks. For Medicare enrollees, surgical MCG care guidelines also identify which procedures CMS has designated as inpatient only.

Use of these evidence-based clinical criteria to support decision making around the need for a given procedure is of benefit to the patient, as all procedures come with inherent risk that must be balanced by anticipated clinical benefit. Utilizing evidence-based clinical criteria enables a more accurate and patient-specific decision-making process. In addition, the use of evidence-based guidelines can help reduce unwarranted variation in care, such as divergent clinical thresholds to perform a procedure for clinically similar patients that vary across geographic regions, between facilities, and among individual clinicians.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(71); 42 CFR 419.22(72); 42 CFR 422.101(73)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(74); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(75); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(76); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(77)

Medicare Coverage Determinations: Medicare Coverage Database(78)

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Footnotes

[A] Treated and cured cancers, including any carcinoma in situ, nonmetastatic basal cell and squamous cell carcinoma of the skin, melanoma in situ, prostate cancer with Gleason score 6 or less, small (3 cm or less) renal tumors, superficial bladder cancer, and follicular or papillary thyroid cancer less than 2 cm with low-grade histology, are not contraindications to transplant. Patients with hematologic or solid tumor malignancies in remission and patients with myelodysplasias, chronic leukemia, or low-grade lymphomas may be transplant candidates after evaluation by a transplant hematologist.(1) [A in Context Link 1]

[B] Assessment of medical risk may include evaluation for recurrent kidney disease, especially systemic amyloidosis, light-chain deposition disease, membranous or membranoproliferative glomerulonephropathy, and hemolytic uremic syndrome; heart disease; pulmonary arterial hypertension; cerebral or peripheral vascular disease; infection; cancer; liver disease; gastrointestinal disease; diabetes; coagulopathy or hypercoagulable state; and age-related comorbidities and risks.(1)(2)(11)(12) [B in Context Link 1]

[C] Indications for native nephrectomy may include polycystic kidney disease with severe recurrent complications (eg, bleeding, infection, stones), insufficient space for the transplant kidney, heavy proteinuria associated with hypoalbuminemia, or if the disease affecting the native kidney predisposes to increased cancer risk.(1) [C in Context Link 1, 2]

[D] See Clinical Indications for Procedure in this guideline. [D in Context Link 1]

[E] Use Multimodal analgesia or individual analgesic agent as indicated. [E in Context Link 1]

[F] Patient will be discharged on immunosuppressive medication and corticosteroids. [F in Context Link 1]

[G] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [G in Context Link 1]

[H] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(49) [H in Context Link 1]

Definitions

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

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Footnotes

- Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP.

Multimodal analgesia

- Multimodal analgesia involves the utilization of 2 or more analgesic agents with different mechanisms of action in order to provide additive or synergistic pain control, while minimizing side effects and reliance on opioids.(1)(2)(3)

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Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Readmission Risk Assessment

- Risk of readmission is increased by presence of **1 or more** of the following(1)(2)(3)(4)(5)(6)(7):
 - Hospitalization (nonelective) in past 6 months
 - Two or more emergency department visits in past 6 months
 - No source of outpatient care other than emergency department (eg, no primary care provider)
 - Severe care transition barriers (eg, no caregiver, homeless)
 - Delayed graft function (eg, need for dialysis within 7 days following transplant)
 - History of cerebrovascular accident (stroke)
 - Prior renal transplant (retransplant needed)

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Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following(7):
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
 - Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
 - Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
 - Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
 - Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Codes

ICD-10 Diagnosis: I12.0, N18.4, N18.5, N18.6, Q61.19, Q61.2, Q61.3, Q61.5, T86.12

ICD-10 Procedure: 0TY00Z0, 0TY00Z1, 0TY00Z2, 0TY10Z0, 0TY10Z1, 0TY10Z2

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